

EEG Cognitive State Machine: A Modular Framework for Design-Time Experimentation and Run-Time Neurofeedback in Puzzle Games

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Abstract—In this proof-of-concept study, we present a subject-tailored EEG Cognitive State Machine (CSM), a reusable brain-computer interface (BCI) layer that exposes EEG-derived cognitive and affective estimates to downstream game software. The framework supports two complementary roles for internal state inference: (i) design-time evaluation for user experimentation, in which inferred state is exposed as an auxiliary metric for comparing game variants, and (ii) run-time closed-loop neurofeedback, in which inferred state is used as a control signal for dynamic adaptation. We instantiate this framework in two game settings. In a virtual reality (VR) spatial cognition task, a BiLSTM regressor predicts user performance with $R^2 \approx 0.6$, serving as a proxy metric to compare difficulty progression strategies in a single-subject longitudinal A/B-style evaluation. In a 2D Tetris game, an EEG valence-arousal BiLSTM classifier drives dynamic difficulty adjustment toward an enjoyment target, showing a suggestive within-subject increase ($d = 1.109$) in reported enjoyment frequency over a game heuristic controller ($N = 4$).

I. INTRODUCTION

Puzzle games and other cognitively demanding interactive systems increasingly rely on run-time adaptation and design iteration to provide optimal experiences. However, most systems remain “externally aware,” inferring the user state from behavioral telemetry such as score, errors, completion time, or controller input. Although these signals are useful, they can be incomplete proxies for the internal states that matter most to user experience, that is, cognitive and affective states. With puzzle games in particular, similar outward behavior can reflect very different internal conditions, making it difficult to evaluate game variants or adapt difficulty with precision.

Recent improvements in consumer electroencephalography (EEG) and machine learning (ML) make it increasingly practical to incorporate internal state sensing into deployable interactive systems. These advances suggest that portable consumer-grade EEG devices can support ML-based estimation of states such as mental workload, engagement, attention, and emotions [1], [2], [3]. In this work, we present a subject-tailored EEG framework for puzzle games that operationalizes inferred cognitive and affective state as a software-level signal for downstream applications. We refer to this integration pattern as a Cognitive State Machine

(CSM), a modular pipeline that acquires EEG during gameplay, performs preprocessing and ML-based inference, and exposes the inferred state to downstream software. The “State Machine” aspect of this framework refers to the addition of cognitive and emotional states to induce computational transitions, building on traditional state machines that use only symbolic states.

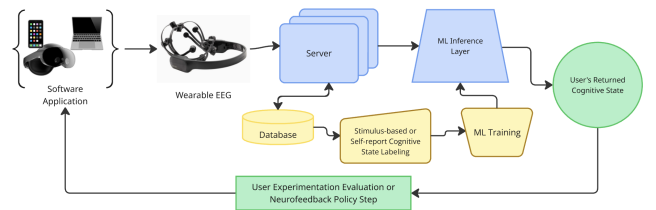


Fig. 1. Flow chart for training/evaluation and integration with downstream VR or desktop/mobile applications.

User Experimentation. While Online Controlled Experimentation (OCE) is standard in consumer software, it is difficult to apply to cognitively demanding games where sample sizes are small and behavioral telemetry acts as an incomplete proxy for internal user states [4], [5]. Identical behavior can reflect different cognitive conditions, and short sessions may hide meaningful experiential differences. A cognitive state-aware experimentation layer addresses this by providing neurophysiological signals that can be compared across variants even when behavioral outcomes are sparse. This augments experiments with sensitive internal state measures that help explain *why* a variant succeeds, supporting faster iteration on software design decisions. Previous work has investigated the use of EEG to assess the affective and cognitive dimensions of user experience [6], [7], [8]. However, these approaches have not established the ability to use EEG as a metric in the user experimentation pipeline, nor have they compared EEG-based UX analysis with traditional user experimentation in an evaluative setting such as A/B testing.

Closed-loop Neurofeedback. The same state estimation logic can drive *closed-loop neuroadaptation*, in which the system continuously infers a user’s current state and adjusts content in real time to steer interaction toward desired regions, such as sustained engagement or reduced overload. Prior EEG-based neurofeedback studies have established a conceptual framework that often treats state estimation as a task-specific classifier or controller for dynamic difficulty adjustment (DDA) [9], [10], [11]. Moreover, these models

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of biocybernetic loops often use hand-crafted signal ratios that implement preset adjustments. Fixed indices and hand-coded mappings from physiological signals to cognitive or affective states are often task-dependent and may lack sufficient accuracy and cross-context reliability, motivating the use of validated predictive models [10]. Previous work has used EEG-based ML either to infer discrete difficulty levels without adapting gameplay [12] or to adapt task difficulty through thresholded workload estimates and fixed step changes [13]. However, these approaches do not use an iterative, fine-grained control law that converts ML-derived affect estimates and prediction confidence into bounded scalar difficulty updates at each control cycle.

The novelty of this work lies in the operationalization of EEG-based state inference as a reusable application-facing layer above EEG middleware and data-acquisition APIs, supporting both design-time evaluation and run-time control. We provide and evaluate a modular CSM for cognitive and affective state inference in puzzle games. The CSM supports two complementary roles:

(1) User Experimentation, in which inferred state is used as a proxy metric for comparing design variants. We demonstrate this in a VR spatial cognition game, where EEG-based performance prediction is used to evaluate difficulty-progression variants in an A/B-style setting [14], in which we compare EEG-based estimation to traditional OCE-style evaluation.

(2) Closed-loop Neurofeedback, in which inferred state is used as a control signal for real-time adaptation in a neuroadaptive Tetris game, where an EEG valence–arousal bidirectional long short-term memory (BiLSTM) model drives DDA to steer gameplay toward enjoyment. By adding an online ML classification-to-adjustment scheme, we enable DDA that responds to estimated mental or affective states through a dynamical-systems controller (shown in Section II-B) rather than only to score or error rates. Online neuroadaptation aims to allow data-driven fine-grained control over the experience, giving control over the user’s baseline adjustment error and avoiding evaluation-only periods without change in the experience.

II. METHODOLOGY

Across two experiments, we implement a modular CSM that (i) collects EEG (128 Hz sampling) signals during interaction and synchronizes them with task events, (ii) applies noise reduction and feature extraction using *EEG-Proc*, an open-source Python package for EEG featurization and ML, (iii) performs ML-based inference of cognitive or affective state estimates, and (iv) exposes the inferred state to downstream software as either an *outcome metric* for user experimentation or a *control signal* for closed-loop neurofeedback. The overarching logic of the CSM is formalized as follows:

$$\mathcal{C}(x_{t-w:t}, m_t) \rightarrow (\hat{s}_t, c_t, \tau_t, a_t), \quad (1)$$

where $x_{t-w:t}$ is the EEG window, m_t is synchronized task metadata, $\hat{s}_t$ is the inferred cognitive/affective state

or continuous estimate, c_t is a confidence score, τ_t is the timestamp, and a_t is an optional downstream action or metric. This formulation supports a clear and modular implementation of the CSM, allowing feature extraction strategies, predictive models, target states, uncertainty estimation, and action policies to be substituted across different tasks.

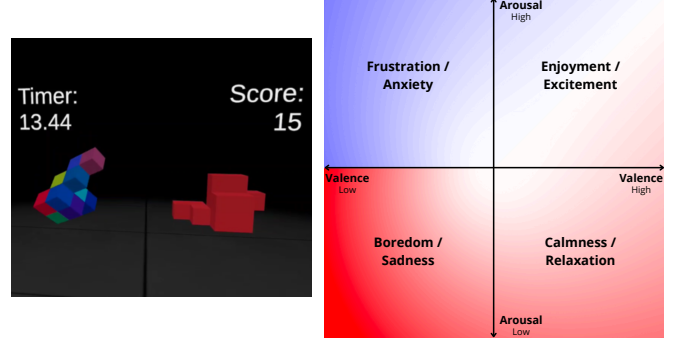


Fig. 2. VR spatial cognition game used in Section II-A, in which the user must rotate the colored object so that it matches the orientation of the red object, alongside the valence–arousal map used in Section II-B.

We extract power spectral density, entropy, and asymmetry features from the θ , α , β_{low} , β_{high} , and γ bands using 2-second windows with 50% overlap before feeding the resulting features into the model. EEG signals were detrended, notch filtered, and bandpass filtered from 4 to 45 Hz using infinite-impulse-response before feature extraction. While these features are prevalent in EEG cognitive/affective prediction literature, we selected them empirically to reduce the feature space, balancing strong predictability with a rapid feature extraction process for real-time inference.

The downstream software can then consume this output either as an evaluation metric, as in the VR A/B analysis, or as a control variable, as in Tetris-EEG. We used the Emotiv EPOC X wireless 14-channel headset for both experiments. In the VR spatial cognition study, the headset was paired with the Meta Quest Pro. To reduce session leakage, we use session-independent validation through leave-K-sessions-out cross-validation (LKOCV).

A. Brain-Enhanced Applications: A/B Testing Difficulty Progressions in a VR Spatial Cognition Game

This experiment tests whether a CSM can make *user experimentation* more informative in VR puzzle and serious games by providing an EEG-derived proxy metric of performance. We implement a VR spatial cognition game and deploy two difficulty progression variants that differ only in how challenge ramps over sessions. In this game, the player rotates a 3D object until it matches the orientation of a target object (Fig. 2). Every time the objects are matched, they merge and spawn a new object to be matched while resetting the timer and adding one point to the score. The participant in this experiment is randomly assigned to one progression per session and plays until the timer hits 0.00. This experiment is conducted across multiple sessions, enabling a single-subject longitudinal A/B comparison [14].

We train a BiLSTM performance predictor on EEG data only, and expose its trial-level estimates to the experimentation layer as a predicted performance score. We divide 16 sessions into 4 equal folds, where each fold has 8 training sessions, 4 validation sessions, and 4 evaluation sessions for reporting. The loss function is modeled using the target performance metric, perf_i , where lower is better:

$$\text{perf}_i = \frac{T_{e,i}}{\left(\frac{E_{\text{Rot},i}}{E_{\text{Rot,max}}}\right) + \left(\frac{N_{\text{Blocks},i}}{N_{\text{Blocks,max}}}\right)}, \quad (2)$$

$$\mathcal{L}_{\text{MAE}} = \frac{1}{|\mathcal{V}|} \sum_{i \in \mathcal{V}} |\widehat{\text{perf}}_i - \text{perf}_i|. \quad (3)$$

Equation 2 defines the trial-level performance score used as the regression target. Then, Equation 3 defines the validation objective, mean absolute error (MAE), over the validation set $\mathcal{V}$. The target performance score is defined as elapsed completion time divided by a normalized trial-difficulty term composed of angular disparity and object size. Specifically, $E_{\text{Rot},i}$ captures the angular disparity of trial i , while $N_{\text{Blocks},i}$ captures the object size in number of blocks. Both are then normalized by their maximum values across all trials. In this way, perf_i represents completion time per unit of normalized task difficulty. Therefore, lower values indicate more efficient task completion, since the participant completed the trial quickly relative to its angular and structural difficulty [15]. The gameplay task elicits spatial reasoning under time pressure, whereas the performance metric summarizes the player’s efficiency. This separation is useful because the target metric is not the same as the elicited behavior of the player. The task is expected to engage several learning-relevant processes, including sustained attention, visuospatial representation, and cognitive workload, which are commonly implicated in spatial cognition and mental rotation tasks [16], [17]. Therefore, we interpret the EEG-predicted performance score as a proxy for task-relevant cognitive efficiency rather than as a direct measurement of any single cognitive state.

We achieve an evaluation MAE ≈ 1.3 (perf_i ranges from 1.9 to 16.1) and $R^2 \approx 0.6$ with LKOCV, suggesting within-subject predictive feasibility for performance in spatial cognition gameplay. It is important to note that the *task* of solving spatial cognition trials is distinct from the *performance metric* used for experimentation. The best resulting hyperparameter configuration of the BiLSTM model was 1024 recurrent units (single bidirectional layer), 256 dense units (2 dense ReLU layers), 0.4 dropout rate (2 dropout layers between bidirectional and dense layers), 0.0001 learning rate (Adam optimizer), 128 batch size, and an early-stopping patience of 50 epochs.

Rather than evaluating only trial-level prediction error, we evaluated whether the EEG-based regressor preserved the decision-relevant temporal pattern of the A/B experiment. To account for unequal session lengths, each trial was indexed by normalized session progress and assigned to one of 10 equal-width bins. Within each bin, we computed the mean true and EEG-predicted performance separately for

Variant A, which maintained fixed difficulty, and Variant B, which progressively increased object size and reduced target time. The A/B effect was then defined as $\Delta_{AB} = \text{perf}_B - \text{perf}_A$, producing corresponding 10-bin curves for the true and predicted values. Because lower perf_i indicates better performance, $\Delta_{AB} < 0$ favors Variant B and $\Delta_{AB} > 0$ favors Variant A. The predicted and true delta curves were strongly correlated ($r = 0.856$, $p < 0.01$), and their signs matched in every bin, indicating that the EEG-derived metric preserved both which variant performed better and how the difference evolved over the session. However, correlation reflects similarity in shape rather than numerical agreement. Bland–Altman analysis showed that predicted deltas were higher on average than true deltas, with a mean difference of 0.588 and 95% limits of agreement from -0.560 to 1.735 . Therefore, a predicted delta of a given magnitude should not be interpreted as the same numerical change in the behavioral performance measure. Instead, the EEG-derived proxy is most appropriately interpreted as preserving the sign, relative pattern, and temporal progression of the A/B effect. Although its systematic bias and bin-level variability prevent it from serving as a calibrated numerical substitute for the behavioral measure, they do not preclude its use as a directional metric for comparative evaluation.

B. DDA Tetris Based on Affective State Classification to Maximize Enjoyment

In a second experiment, we evaluate the implementation of a CSM as a practical *closed-loop neurofeedback* layer for games by driving DDA from inferred affect rather than from score-only telemetry. Concretely, we implement a real-time EEG–ML pipeline embedded in an open-source Tetris implementation that adapts game difficulty to keep players near an *enjoyment* region in the valence–arousal circumplex (high valence, high arousal). Although EEG-based neurofeedback for Tetris is not new, prior work has largely focused on other control targets and adaptation schemes, including early biocybernetic game adaptation based on cognitive-state measures [11] and affect-driven selection among a small set of discrete game speeds [18]. In contrast, our implementation uses ML-based valence–arousal prediction to drive continuous difficulty adaptation toward a target affective state, rather than relying on handcrafted band-ratio control logic or selecting among a small set of predefined speeds. This experiment instantiates the CSM as a run-time controller, in which inferred affective state is not used as an outcome measure after the session but as an online signal that directly updates game difficulty.

We first implement classic Tetris, defining difficulty as the drop speed in milliseconds and removing confounding commands that can change the game’s pace (e.g., hard drop and soft drop). We apply three difficulty modules: (a) a simple baseline that increases difficulty with progress (lines cleared), (b) an advanced game heuristic controller that incorporates board-state features such as isolated cells and pieces placed, and (c) a CSM controller that drives the user toward *enjoyment* based only on affective state classification.

We collect EEG data of gameplay across four participants ($N = 4$) and stream features through a backend server.

Every two minutes, participants are notified and required to report their emotional-state quadrant aligned with Russell’s circumplex (relaxation, boredom, anxiety, enjoyment) [19]. Participants were also allowed to report if they felt another emotion during the 2-minute interval. If a transition between emotions occurred, the interval was split in two. The reported emotion quadrants are converted into a valence–arousal coordinate (v, a) , where each axis is discretized into binary classes, and propagated to every 2-second EEG window within the 2-minute interval. The 2-minute interval was chosen based on our interpretation of previous studies on the fluctuation time of emotions and on the overlapping transient nature of emotions [20], [21]. Another reason for 2-minute self-report intervals is to avoid frequent disruptions to the user experience. There are limitations to this, however, such as end-of-window recency bias, changes in participants’ reporting criteria due to fatigue or emotional baseline adjustment, and overlooked rapid emotional fluctuations.

To collect our EEG–Tetris dataset, we first ran 30-minute trials with difficulty module (a) to create an initial dataset and train a BiLSTM on it. Then, we ran 30-minute trials on (b) and (c) (assigned randomly) to evaluate which version promoted increased enjoyment. At run-time, the model predicts valence and arousal from sliding EEG windows and maps them into an emotional quadrant. The state classifications are then used by the EEG-driven DDA controller, difficulty module (c), to update the drop speed by Δs milliseconds.

The EEG-adaptive controller estimates affect over 2-second sliding EEG windows and computes an emotional distance. The predicted state is measured against a neutral reference point (m_x, m_y) : $x = c_t(v - m_x)$, $y = c_t(a - m_y)$, where c_t is confidence. This offset is converted to a radius $r = \sqrt{x^2 + y^2}$, and an angle $\theta = \text{atan2}(y, x)$. Lastly, ϕ_1 is the target direction, pointing toward the center of the high-valence, high-arousal quadrant. These are used in the Emotional Distance Minimizer (EDM) difficulty function:

$$\Delta s = -\tanh(kr) \frac{r}{R_{\max}} \frac{\theta - \phi_1}{\alpha}, \quad (4)$$

where k is a proportional controller constant, $R_{\max}$ is the maximum radius of the circle, and α is a constant regularizer.

Δs is a regularized update, in which its magnitude scales with how far the predicted state sits from the neutral point *and* with how far its direction is from the enjoyment target, so the two distances jointly control the response. Intuitively, when the player is estimated to be bored or relaxed, the controller increases Δs to raise the challenge. When the player is estimated to be anxious, the controller decreases Δs to reduce overload.

We evaluate the feasibility of real-time inference, achieving < 0.5 s total processing time per spawned block (split between feature extraction, prediction, and the Δs function). We split the data at the session level for subject-dependent cross-validation. We hold out $k = 5$ sessions from each user for the validation set that is used for model selection. The

selected BiLSTM model achieved 77% ($\sim 63\%$ baseline) arousal accuracy and $\sim 60\%$ valence accuracy ($\sim 52\%$ baseline) in LKOCV. The best resulting hyperparameter configuration of the BiLSTM model was 512 recurrent units (single bidirectional layer), 256 dense units (2 dense ReLU layers), 0.2 dropout rate (2 dropout layers between bidirectional and dense layers), 0.0001 learning rate (Adam optimizer), 128 batch size, and an early-stopping patience of 50 epochs.

The EEG-adaptive controller showed a 31% relative increase in enjoyment over the game heuristic controller, although the exploratory cross-subject ANOVA was not significant ($p = 0.246$) and was severely underpowered given the small sample. We therefore emphasize the within-subject analysis calculated using proportion z-tests at the self-report interval level. Cohen’s d shows the EEG-adaptive controller produced higher enjoyment than both the baseline condition ($d = 1.376$, +21.2 percentage points) and game heuristic condition ($d = 1.109$, +17.1 percentage points), whereas the baseline and game heuristic conditions differed minimally ($d = 0.222$). Responses varied across participants: enjoyment increased from 42.0% to 49.7% for participant 1 ($p = 0.1121$), decreased slightly from 39.4% to 37.3% for participant 2 ($p = 0.7189$), increased substantially from 9.2% to 62.8% for participant 3 ($p < 0.0001$), and increased from 51.7% to 60.9% for participant 4 ($p = 0.1039$). Although participants 1 and 4 showed suggestive increases, the aggregate effect was driven primarily by participant 3. Nevertheless, all four participants reported greater enjoyment under the EEG-adaptive controller than under the baseline condition. Given the small sample and repeated within-participant observations, these comparisons are interpreted as exploratory rather than population-level evidence.

III. DISCUSSION

This paper provides a basis for integrating EEG-based BCI methods with existing platforms, including extended reality and mobile/desktop interfaces, by exposing subject-tailored EEG state estimates through a reusable software interface for both OCE and neuroadaptation. In both experiments, the key benefit of the proposed CSM is that it makes internal cognitive/affective state a first-class signal for both evaluation and control. In Section II-A, we use the CSM to provide an evaluation strategy that can differentiate difficulty progressions even when behavioral telemetry is sparse or ambiguous. In Section II-B, the CSM becomes a real-time controller through an EEG-driven DDA loop that attempts to regulate affect toward enjoyment, achieving feasible latency (< 0.5 s per spawned block) and a suggestive increase in enjoyment frequency in within-subject evaluation ($d = 1.109$, +17.1 percentage points). Taken together, these results support the feasibility of reusing a cognitive state inference layer across applications as both an auxiliary evaluation signal and a run-time control signal.

Although physiological information is only one component of UX measurement, EEG can provide an additional neurophysiological signal that is not always visible in behavioral telemetry. CSM-based experimentation may therefore

improve interpretability by showing whether a design variant is associated with changes in estimated cognitive or affective state, especially when behavioral outcomes are sparse, delayed, or ambiguous. This methodology may be useful for testing and enhancing games, extended reality, and e-learning platforms by adding a cognitive/affective measurement layer to conventional interaction logs. The same state-estimation layer also creates a pathway toward personalized immersive experiences by allowing interfaces to adapt based on estimated user state.

These results suggest that subject-tailored EEG-based state estimates may provide an auxiliary signal for design-time comparison and run-time adaptation, but the current studies should be interpreted as feasibility demonstrations rather than population-level evidence. The experiments, therefore, emphasize the CSM as a shared layer for both OCE-style evaluation and live neuroadaptive control. Integrating non-invasive BCIs into deployable software environments helps connect physiological and affective computing to user-facing products by treating estimated state as an operational metric and a control variable inside adaptive software.

A major limitation of the current approach is generalization across users, tasks, and labeling contexts. Published EEG cognitive and affective datasets are useful for benchmarking, but task mismatch and self-report inconsistency can limit direct transfer to interactive gameplay. The current results are therefore intentionally proof-of-concept and largely personalized, where Section II-A is single-subject longitudinal and Section II-B's small sample size limits generalizability. Future work should focus on *generalization* through prolonged multi-user studies across diverse tasks, transfer learning across datasets and application contexts [22], [23], and uncertainty-aware EEG classifiers that can be jointly optimized with neuroadaptive controllers. These directions are critical for reusing a CSM across multiple software applications without recreating state labeling and ML modeling for every application category.

CODE AVAILABILITY

The implementations used in this study are available at <https://github.com/EEGo-UNC/EEGProc> and <https://github.com/EEGo-UNC/Tetris-EEG>.

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